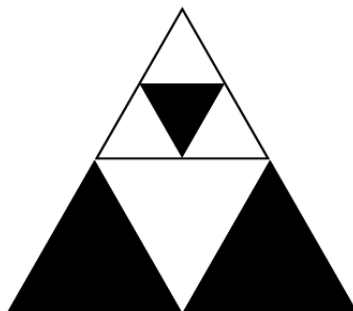


- If $p = 7$ and $q = 2(p - 5)$, compute $q(p + 5) - 2$.
(A) 286 **(B)** 142 **(C)** 100 **(D)** 61 **(E)** 46
- The operation $\S$ is defined by $x\S y = x^2 + xy + y^2$. Find $(1\S 2)\S 3$.
(A) 49 **(B)** 58 **(C)** 70 **(D)** 79 **(E)** 91
- In the figure shown, an equilateral triangle is split into four congruent smaller triangles, one of which is split again in the same way. What fraction of the figure is shaded?



- (A)** $\frac{1}{2}$ **(B)** $\frac{13}{24}$ **(C)** $\frac{5}{9}$ **(D)** $\frac{9}{16}$ **(E)** $\frac{7}{12}$
- Compute $2021 + \frac{2023! + 2024!}{2022! \times 2025}$, where $n! = 1 \times 2 \times \dots \times (n - 1) \times n$.
(A) 4044 **(B)** $\frac{4044}{2023}$ **(C)** $\frac{4088484}{2023}$ **(D)** $\frac{4094552}{2023}$ **(E)** 2023
 - Cards are drawn one at a time from a deck containing 3 red cards and 2 black cards. What is the probability that all the black cards are drawn before all the red cards are drawn?
(A) $\frac{2}{5}$ **(B)** $\frac{1}{10}$ **(C)** $\frac{7}{10}$ **(D)** $\frac{3}{5}$ **(E)** $\frac{2}{3}$
 - Positive numbers x, y, a, b satisfy the inequalities $x < y$ and $a < b$. How many of the following inequalities are necessarily true?
 (i) $x + a - b < y - a + b$
 (ii) $\frac{x + y}{b} < \frac{x + y}{a}$
 (iii) $\frac{x}{b} < \frac{y}{a}$
 (iv) $a - y < b - x$
 (v) $ax < by$
(A) 1 **(B)** 2 **(C)** 3 **(D)** 4 **(E)** 5
 - The perpendicular bisector of a radius of a circle intersects the circle at points A and B. Given that $AB = 12$, find the radius of the circle.
(A) $2\sqrt{3}$ **(B)** 6 **(C)** $4\sqrt{3}$ **(D)** 8 **(E)** $6\sqrt{2}$
 - How many possible values of a are there such that $x^2 + ax + 36$ divides the product $x(x + 1)\dots(x + 35)(x + 36)$?
(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 12

9. Daniel has x dollars and Elaine has y dollars. Elaine gives Daniel a **fifth** of her money, which gets added to his bank account. Daniel then gives Elaine a **third** of his total money, which gets added to her bank account. Finally, Elaine gives Daniel a **quarter** of her money, after which she notices that they both have the same amount of money. Find the ratio $\frac{x}{y}$.

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{5}$ (D) $\frac{4}{7}$ (E) $\frac{5}{7}$

10. A game of whack-a-mole consists of one mole and eight holes. On each turn, the player chooses a hole to whack randomly and the mole chooses a hole to pop out of randomly. If the player plays for seventeen turns, then $\frac{m}{n}$ is the average number of times they will hit the mole, where m, n are co-prime integers. What is $m + 10n$?

(A) 87 (B) 178 (C) 12 (D) 78 (E) 97

11. A function is defined recursively with $f(0) = 1$ and $f(x) = (2x - 1)f(x - 1)$. Find the smallest integer n such that $f(n)$ is divisible by 2023^2 .

(A) 49 (B) 60 (C) 73 (D) 91 (E) 119

12. A sequence is defined with $a_1 = 2, a_2 = 3$, and $a_n = a_{n-1}a_{n-2} - 1$. Find the last digit of a_{2023} .

(A) 7 (B) 9 (C) 4 (D) 2 (E) 5

13. Evaluate

$$\left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{9}\right) \left(1 - \frac{1}{16}\right) \cdots \left(1 - \frac{1}{2025}\right).$$

(A) $\frac{23}{45}$ (B) $\frac{4}{2025}$ (C) $\frac{1}{2}$ (D) $\frac{1013}{2025}$ (E) $\frac{169}{529}$

14. If $f(x)$ is a quadratic that satisfies $f(1) = f(f(1)) = 2$, compute $f(5) - 6f(3)$.

(A) 6 (B) 2 (C) -2 (D) -10 (E) 0

15. For how many bases b , with $3 < b < 2023$, is 2023_b divisible by 7?

(A) 576 (B) 578 (C) 866 (D) 867 (E) 1156

16. Inside of square $ABCD$ with side length 3, a circle ω with radius 1 is drawn such that it is tangent to both $\overline{AB}$ and $\overline{BC}$. Point G lies on $\overline{CD}$ such that $\overline{AG}$ is tangent to ω . Find the area of triangle AGD .

(A) $\frac{27}{8}$ (B) 3 (C) 4 (D) $\frac{7}{2}$ (E) $\frac{15}{4}$

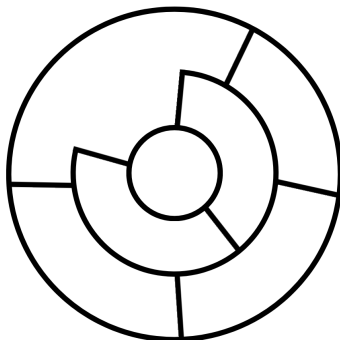
17. Avogadro, Bernoulli, Cauchy, Darwin, Einstein, Fermat, and Galileo are lining up to enter the exam room for OMC. How many ways are there for them to do so if Avogadro cannot be first in line, Cauchy cannot be 3rd in line, Einstein cannot be 5th in line, and Galileo cannot be last in line?

(A) 2790 (B) 2250 (C) 6 (D) 2880 (E) 2472

18. In triangle ABC , point D lies on $\overline{BC}$ such that $\angle ADB = 90^\circ$. If $\overline{AD} = 5$, $\overline{BD} = 2$, and $\angle BAC = 45^\circ$, find the length of $\overline{DC}$.

(A) 2 (B) $\frac{15}{7}$ (C) 3 (D) $\frac{30}{13}$ (E) $\frac{30}{17}$

19. How many ways can the shown figure be painted with six distinct colours if no two neighboring regions can share the same colour?



- (A) 9720 (B) 12960 (C) 14400 (D) 15480 (E) 17280
20. Player 1 and Player 2 are playing a game. On each turn, a fair coin is flipped. If it lands heads, Player 1 moves forward one square. If it lands tails, Player 2 moves forward one square. A player wins when they are 5 squares in front of the other player. If Player 1 is currently 1 square in front of Player 2, then the probability that Player 1 wins is $\frac{a}{b}$, where a, b are co-prime integers. What is $2a + 3b$?
- (A) 21 (B) 19 (C) 8 (D) 17 (E) 23
21. In equilateral triangle ABC , $\overline{AB} = 6$. Consider a point P which satisfies $\overline{AP} = 3$. Let Q lie on PC such that $CQ : QP = 1 : 2$. The minimum possible length of $\overline{BQ}$ can be written in the form $a\sqrt{b} + c$, where a, b, c are integers and b is not divisible by the square of any prime. Find $a + b + c$.
- (A) 11 (B) 12 (C) 8 (D) 10 (E) 14
22. The largest real solution to $x^4 + 2x^3 - 1686x^2 - 6x + 9 = 0$ can be expressed as $a + \sqrt{b}$, where b is a positive integer. What is $10a + b$?
- (A) 228 (B) 234 (C) 597 (D) 603 (E) 2012
23. How many ways are there to completely tile a 3×10 grid with 1×2 rectangles?
- (A) 418 (B) 243 (C) 162 (D) 3888 (E) 571
24. Parallelogram $ABCD$ has $\overline{AB} = 17$ and $\overline{BC} = 10$. Two circles with diameters AB and BC respectively are drawn, which intersect each other at points B and P with $\overline{BP} = 8$. If the two diagonals of $ABCD$ have lengths x and y , find $|x^2 - y^2|$.
- (A) 266 (B) 538 (C) 104 (D) 310 (E) 907
25. The minimum value of

$$\sqrt{x^2 + y^2} + \sqrt{x^2 + y^2 + 14x + 49} + \sqrt{x^2 + y^2 - 16y + 64},$$

where x and y are real numbers, is z . If $z^2 = a + b\sqrt{c}$, where c is not divisible by the square of any prime, find $a + b + c$.

- (A) 200 (B) 128 (C) 256 (D) 131 (E) 172